Kazi Reyazul Hasan

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Research Focus

- Language Processing
- Speech & Gesture
- Behavioral Modeling
- Multimodal AI

Education

B.Sc. in Computer Science and Engineering Feb 2020 – Mar 2025
Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
CGPA: 3.72/4.00 (last 4 semesters: 3.88/4.00; 2× Dean's List)
Thesis: *MediaPipe-Aided Ensemble Learning with Attention-Based Transformers for Word-Level Sign Language Recognition*

Research Experience

Research Assistant Dec 2023 – Present
BUET – Supervisor: Dr. Muhammad Abdullah Adnan Dhaka, Bangladesh

- VISION **EMPATH** (ICPR 2024, RRPR 2024; both oral). Built a word-level Bangla sign language recognizer that predicts MediaPipe keypoint gaps during fast hand motion with frame interpolation and wrist alignment, then feeds a spatial encoder with a temporal transformer. Improved prior results by 5–8% across INCLUDE, SignBD-Word and MSL | Code | Slides
- AUDIO **Dual-Stream DNN-KAN** (Interspeech 2026). Combined a deep neural network with a Kolmogorov–Arnold Network over Bangla-specific acoustic and phonetic features, reaching 95.35% accuracy on SUBESCO with only 1.1M parameters, comparable to far larger models such as wav2vec2 | Code
- TEXT **BLP 2025 Shared Task** (IJCNLP-AACL 2025 Workshop). Used TF-IDF-informed prompt engineering with LoRA fine-tuning for Bangla hate speech detection.

Research Assistant Oct 2025 – Present
BUET – Supervisor: Dr. A. B. M. Alim Al Islam Dhaka, Bangladesh

- VISION **BanglaLRS** (WACV 2027). Built the first continuous, open-vocabulary Bengali visual speech recognition benchmark: 100 hours of speaker-disjoint news video. A fine-tuned Conformer (auto-avsr) baseline reaches 35.7% CER / 56.5% WER, versus 52% CER on the 220-hour Persian Arman-AV | Code

Research Assistant (Joint Projects) 2025 – Present
BUET – Co-supervised by Dr. Adnan and Dr. Alim Al Islam Dhaka, Bangladesh

- TEXT **Training-Free Writing Biometrics** (IJCB 2026). Proposed an authorship pipeline that uses LLM-generated stylometric profiles over frozen sentence encoders, supporting both verification and identification without task-specific training.
- TEXT **BanglaSTEM** (ACL SRW 2026). Created the first technical-domain Bangla–English parallel corpus, with 5,000+ sentence pairs across mathematics, physics and programming, plus a term-weighted metric that scores how well specialized terminology survives translation | Model

Research Mentor Jun 2026 – Present
BUET – Mentoring Md. Asif Ali and Md. Mosharaf Hossain Apurbo Dhaka, Bangladesh

- VISION **Cross-Signer Sign Recognition** (WSLP, EMNLP 2026). Guided a metric-learning retrieval system for Indian Sign Language where most words have only one training video. Pretrained on a multi-signer ASL corpus, then fine-tuned on the target vocabulary, reaching 67.11% Top-1 on the blind test set.
- VISION **Pose-Based Sign Spotting** (WSLP, EMNLP 2026). Guided a 2.86M-parameter Siamese pose encoder with MIL-NCE contrastive training, reaching 71.95 macro-F1 on the blind test set versus 58.42 for the official pose baseline.

Publications

Peer-Reviewed Conference Papers

[1] **BanglaLRS: A Continuous, Open-Vocabulary Benchmark for Bangla Visual Speech Recognition**
Kazi Reyazul Hasan, Chowdhury Mohammad Rakin Haider, and A. B. M. Alim Al Islam
IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV 2027**) | accepted, to appear

[2] **Toward Training-Free Writing Biometrics: LLM Stylometric Profiles for Authorship Verification and Identification**
Kazi Reyazul Hasan, A. B. M. Alim Al Islam, and Muhammad Abdullah Adnan
International Joint Conference on Biometrics (**IJCB 2026**) | accepted

[3] **Dual-Stream DNN-KAN Networks with Bangla-Specific Features for Speech Emotion Recognition**
Kazi Reyazul Hasan and Muhammad Abdullah Adnan
Interspeech 2026, Main Track | Paper

[4] **EMPATH: MediaPipe-Aided Ensemble Learning with Attention-Based Transformers for Accurate Recognition of Bangla Word-Level Sign Language**
Kazi Reyazul Hasan and Muhammad Abdullah Adnan
International Conference on Pattern Recognition (**ICPR 2024**), Springer | *Oral presentation* | Paper

Workshop Papers (* corresponding author)

[5] **Metric-Learning Retrieval for Cross-Signer Isolated Sign Language Recognition**
Md. Asif Ali, Md. Mosharaf Hossain Apurbo, and **Kazi Reyazul Hasan***
Workshop on Sign Language Processing (**WSLP**), **EMNLP 2026** | Paper

[6] **Multiple-Instance Contrastive Alignment for Pose-Based Sign Spotting**
Md. Mosharaf Hossain Apurbo, Md. Asif Ali, and **Kazi Reyazul Hasan***
Workshop on Sign Language Processing (**WSLP**), **EMNLP 2026** | Paper

[7] **BanglaSTEM: A Parallel Corpus and Term-Weighted Evaluation for Technical Bangla-English Translation**
Kazi Reyazul Hasan, A. B. M. Alim Al Islam, and Muhammad Abdullah Adnan
ACL Student Research Workshop (**ACL SRW 2026**) | Paper

[8] **TF-IDF Informed Prompt Engineering with LoRA Fine-tuning for Bangla Hate Speech Detection**
Kazi Reyazul Hasan, Mubasshira Musarrat, and Muhammad Abdullah Adnan
Shared Task Paper, 2nd BLP Workshop, **IJCNLP-AACL 2025** | Paper

[9] **Enhancing Word-Level Sign Language Recognition**
Kazi Reyazul Hasan and Muhammad Abdullah Adnan
Reproducible Research in Pattern Recognition (**RRPR 2024**), ICPR Workshop, Springer | *Oral presentation* | Paper

Ongoing Works

- **Script Similarity and Data Mixture Inference**, *Under review*, ACL Rolling Review (**ARR**), October 2026 | Tests if tokenizer-based training-mixture inference confuses same-script languages.
- **Text-Mediated Face–Voice Association**, *In preparation* for **ICML 2027** | Tests if a shared text space can carry identity-level features to link faces and voices zero-shot.

Professional Experience

Software Engineer, R&D Jul 2025 – Present
TigerIT Bangladesh Limited – Team Lead: Nasir Ahmed Dhaka, Bangladesh

- Ranked **1st** on the FVC-onGoing Differential Morphing Attack Detection benchmark with an AdaFace ResNet backbone and a lightweight head: 0.00% EER (MORPHDB_D-1.0), 0.19% EER (MORPHDB_P&S-1.0).
- Developing circular-CNN iris recognition, robust to poor lighting, rotation and occlusion, to replace Hamming-distance matching. Current system ranks **14th globally on NIST IREX 10**.
- Built multi-camera warehouse person tracking that keeps identities consistent across camera changes.

Adjunct Lecturer (Part-time) Jan 2026 – Present
Presidency University Dhaka, Bangladesh

- Taught 200+ students across Introduction to Computer Programming, Object Oriented Programming, Digital Logic Design, and Computer Networks and Security.

Junior AI Engineer (Remote) Mar 2025 – May 2025
Robofication LLC Michigan, United States

- Designed a system-modeling interface with JointJS, FastAPI and React for SysML diagram generation.
- Developed LLM-based strategies for exchanging artifacts between a proprietary AI toolchain and Cameo Systems Modeler.

Academic Service

2026 **Program Committee Member**, Reproducible Research in Pattern Recognition (RRPR 2026), ICPR Workshop

2026 **Reviewer**, AIMS Workshop, CVPR 2026

Grants & Awards

Nov 2026 SPARRSO Research Fellowship (FY 2026–27), SAR Fusion and Super Resolution Group, Bangladesh Space Research and Remote Sensing Organization

Nov 2026 Broadening Participation Travel Award, WACV 2027

Aug 2026 CodeCrafters-Investortools Research Grant

Jul 2026 Travel Grant, Interspeech 2026, Sydney, Australia

Aug 2025 SignSpeak selected for the BUET Incubator Program

Sep 2024 Research Grant, ICT Innovation Fund, Government of Bangladesh

Apr 2024 Internal Research Grant, BUET Research and Innovation Centre for Science and Engineering

Mar 2023 Best Project Award, CSE-316 (Microprocessors, Microcontrollers and Embedded Systems), BUET

Test Scores

TOEFL iBT **108/120** (Reading 27, Listening 30, Speaking 24, Writing 27) | Sep 2025

Open Resources & Projects

BanglaSTEM-T5 *Hugging Face Model* | Model-Stats | 350+ downloads
Technical-domain Bangla–English translation model released with BanglaSTEM (ACL SRW 2026).

Skincare Product DBs *Kaggle Datasets* | Skinsort (19K+) | SkinSAFE (50K+) | 1,100+ downloads
Scraped and cleaned product databases with ingredient-level details for recommendation research.

BusBuddy *Node.js, Svelte, PostgreSQL, AWS* | Code
Led development of the admin interface for BUET's bus automation service.

Race *C++, Arduino* | Code | Award
Hardware racing game with two custom-built microcontroller cars.

Leadership & Activities

- Qualified 1st nationally to represent Bangladesh at the International Astronomy Olympiad (IAO 2018).
- Academic Team Member, Bangladesh Mathematical Olympiad (5+ years).
- Joint Secretary and other leadership roles across multiple student organizations; 50+ awards in olympiads, quizzes and debate competitions.

Technical Skills

Deep Learning PyTorch, TensorFlow/Keras, Hugging Face (Transformers, Hub), scikit-learn

Vision & Speech OpenCV, MediaPipe, ViT, Conformer (auto-avsr), AdaFace, wav2vec2, Kolmogorov–Arnold Networks

NLP & LLMs T5 fine-tuning, LoRA, prompt engineering, sentence encoders and embeddings, LangChain

Systems & Web FastAPI, React, Node.js, Svelte, JointJS, PostgreSQL, Firebase, Docker, Git, AWS, Azure, Arduino

Languages Python, C++, Java, JavaScript